

From the labs

Nature runs on its own complex design. Can we emulate them to technology in our daily lives?



Waste to jet fuel

Airlines are working on new ways to reduce greenhouse gases by converting wood waste alcohol to biofuel. <http://digit.in/WWAtoJF>

EXPERIMENTING AT THE FINAL FRONTIER

The most interesting studies
performed where Newton's apple
won't fall

The International Space Station -
floating around since 1998

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Prepare for lift off

An integral part of the human spirit of conquest is the acquisition of knowledge. We isolate phenomena in the world and meddle with it to deduce descriptions that best fit our understanding. For performing science rigorously the 'isolation' becomes imperative because if there are many forces at play, it becomes harder to be sure of the proposed causal relations. There's always a lot going on Earth, rendering space into an almost ideal playground. Other than understanding the playground itself, space offers the opportunity to perform experiments free from (technically speaking, controlled for) the effects of gravity. Well, almost free, since there's

always microgravity. Other than the escape from gravity, another motivation for these experiments is progress towards our future life in space. Here we present to you some of the coolest experiments performed in space!

Astro-wolverine (Space tissue loss - stem cell regeneration)

This experiment, conducted between March and September of 2010, studied some of the effects of the removal of 'mechanical loading' due to gravity, on the embryonic stem cells of mice. As they can differentiate into any cell of the adult body, the embryoid body is considered a model for studying how cells differentiate and develop. This experiment aimed to study the effects of microgravity on stem cell differentiation and the associated gene expressions of

the process. As Earth's gravity is crucial for the development and regeneration of our cells, long-term disuse or removal from this gravitational environment causes atrophy, both by degeneration of existing tissue as well as impeded stem cell regeneration. All cells come from stem cells, so it was hoped that this experiment would give crucial insights and pave the road for further studies of the regeneration mechanisms of our body, both on Earth and in space.

The embryonic stem cells were specially cultured with a growth factor (LIF – Leukemia Inhibiting Factor) to maintain their immortality. Once this growth factor was removed, the cells began to do in the lab what they would do in the body – turn into the various non-dividing cells of the body. The patterns of gene expression during this process